# Vertical Farming Limitations and Potential Demonstrated by Back-of-the-Envelope Calculations

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## Abstract

Vertical farming is gaining traction as a controlled-environment approach for food production, but its large-scale potential is constrained by fundamental physical limitations. Using back-of-the-envelope calculations, this perspective assesses vertical farming's energy efficiency, cost structure, and environmental footprint. Current commercial systems convert only ~1-2% of electrical energy into plant biomass. LED lighting improvements could reach a theoretical maximum of ~11%, but remain far below natural photosynthesis efficiency in well-managed greenhouses. Economic and environmental analyses suggest vertical farming is best suited for specific high-value crops and contexts where conventional agriculture is constrained.

***Keywords:*** *vertical farming · energy efficiency · controlled environment agriculture · LED lighting · photosynthesis · food systems · sustainability*

## 1. The Promise and Challenge of Vertical Farming

Vertical farming - growing crops in stacked layers under artificial lighting - promises year-round production independent of weather, dramatically reduced water use, no pesticides, and proximity to urban consumers. These attributes have attracted substantial investment and enthusiasm.

Yet for all this promise, vertical farms face a fundamental physical challenge: replacing sunlight with electricity is expensive and thermodynamically inefficient. This perspective uses simple, transparent quantitative estimates - 'back-of-the-envelope' calculations - to establish benchmarks for vertical farming's energy use, costs, and environmental footprint, and to identify where the technology's potential is greatest.

Rather than comprehensive techno-economic modeling (which often obscures assumptions in complexity), we use deliberately simplified calculations to reveal key constraints and opportunities. This approach makes the analysis reproducible and the assumptions explicit.

## 2. Energy Efficiency: Current State and Theoretical Limits

### 2.1 Current Energy-to-Biomass Conversion

To estimate how efficiently vertical farms convert electrical energy into plant biomass, we trace the energy cascade. Starting from 100 units of electrical energy: LED fixtures convert electricity to light at ~50% efficiency; ~85% of LED output falls in the photosynthetically active radiation (PAR) band (400-700 nm); plants absorb ~90% of intercepted PAR; the theoretical maximum quantum efficiency of photosynthesis is ~11%; harvest index (edible fraction) is ~50% for leafy greens.

Multiplying these factors: 0.50 x 0.85 x 0.90 x 0.11 x 0.50 = ~2.1%. Current commercial systems measured empirically achieve 1-2%, matching this estimate. The main losses are in photosynthesis itself - a hard physical limit, not an engineering failure.

### 2.2 Theoretical Maximum with Improved LEDs

Future LED improvements could approach the thermodynamic limit. With 80% efficient LEDs: 0.80 x 0.90 x 0.95 x 0.11 x 0.50 = ~3.8%. Even with optimistic assumptions, the theoretical ceiling is approximately 11% - the photosynthetic quantum efficiency limit.

For comparison, well-managed Dutch greenhouses using supplemental lighting achieve ~8% solar-energy-to-biomass efficiency. Sunlight is effectively free; electricity costs money and carries a carbon footprint. This comparison is fundamental to understanding vertical farming's niche.

## 3. Cost Analysis

### 3.1 Electricity as Dominant Operating Cost

At 1-2% efficiency, producing 1 kg of leafy greens requires ~20-40 kWh. At $0.10/kWh (commercial rate), that is $2-4 of electricity per kilogram - comparable to the wholesale price of commodity lettuce (~$1-3/kg). For high-value crops like herbs (~$10-30/kg wholesale), the energy cost is manageable.

This explains why vertical farms concentrate on high-value, fast-growing leafy crops. Staple crops like wheat or rice would require orders-of-magnitude more energy per calorie than field crops - making large-scale staple production economically untenable under current conditions.

### 3.2 Capital Costs

Vertical farms require substantial upfront capital: climate-controlled buildings, LED arrays, hydroponic systems, and automation. Capital costs of $5-15 million per hectare of growing area are common, versus $10,000-100,000/ha for greenhouses and ~$5,000-50,000/ha for field agriculture. Financial viability requires high land productivity and premium crop prices.

## 4. Environmental Footprint

### 4.1 Carbon Footprint of Electricity

At 20-40 kWh per kg of lettuce, using US average grid intensity of ~0.4 kg $CO_2$/kWh, the carbon footprint is ~8-16 kg $CO_2$ per kg. Compare: field-grown lettuce ~0.5-1 kg $CO_2$/kg; greenhouse lettuce ~1-3 kg $CO_2$/kg.

Grid-powered vertical farms currently have substantially higher carbon footprints. With renewable electricity (0.02-0.05 kg $CO_2$/kWh from solar/wind), the footprint drops to ~0.5-2 kg $CO_2$/kg - competitive with conventional production. Decarbonizing the electricity supply is a necessary condition for vertical farming to offer genuine climate benefits.

### 4.2 Water Use

Vertical farms typically use 95-99% less water than field agriculture through recirculation with no soil evaporation or runoff. This is a genuine advantage in water-scarce regions. However, for most crops and regions, water is not the binding constraint - energy and cost are.

### 4.3 Land Use

Vertical stacking increases yields per unit footprint by 10-100x compared to greenhouses - significant in land-constrained urban settings. However, global agricultural land scarcity is not the primary food security challenge in most regions, and 'saved' land is often not where vertical farms are built.

## 5. Pathways to Improved Efficiency

Research directions that could improve vertical farming efficiency: LED spectral optimization (targeting wavelengths most efficiently used by photosynthesis); crop engineering for higher photosynthetic efficiency and harvest index; improved $CO_2$ enrichment (1,000-1,500 ppm can increase photosynthetic efficiency by 20-40%); thermal management (recovering LED waste heat for space heating).

Even with all these improvements, the ~11% photosynthetic quantum efficiency limit means grid electricity will remain a major cost. Pairing vertical farms with dedicated renewable energy (e.g., solar panels on the building) could simultaneously reduce costs and carbon footprint.

## 6. Where Vertical Farming Makes Most Sense

Vertical farming's strongest case: (1) land is extremely scarce and expensive (dense urban areas, islands, polar regions); (2) crops are high-value and weather-sensitive (herbs, microgreens, specialty leafy greens); (3) electricity is from renewable sources; (4) conventional supply chains are long or unreliable; (5) water is extremely scarce. Large-scale commodity crop production for broad food security is unlikely to be cost-competitive with conventional agriculture under foreseeable technological improvements.

## 7. Outstanding Questions

Key open questions: Can photosynthetic efficiency be engineered beyond current limits? What is the realistic learning curve for vertical farm capital costs? How will AI and robotics change labor economics? Can vertical farms profitably serve specialty pharmaceutical or nutraceutical markets? How do externalities (pesticide avoidance, reduced transport, food safety) factor into true cost comparisons?

## 8. Conclusion

Back-of-the-envelope calculations reveal that vertical farming's 1-2% energy-to-biomass conversion efficiency is not poor engineering, but a consequence of fundamental physical constraints in photosynthesis. Future improvements could push efficiency to 5-10%, but the theoretical ceiling is ~11%. Grid electricity costs will remain high, and vertical farming is best suited for high-value niche crops in specific geographic contexts rather than as a general solution to global food production. Renewable electricity integration is essential for genuine climate benefits.

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